Student ID: _____

Sukkur IBA University

Mid Term Examinations

Spring 2025

Course: Artificial Intelligence (E) | **Class:** BS(CS/SE)-VI

Date: 15th May, 2025 | **Time:** 60 mins

Instructor: Atiqa Aiman | **Marks:** 5 Marks

Instructions:

1. *Name document your name_cmsid.*
2. *Use this document to add output screenshots below each task question.*
3. *Provide code files and output screenshot*
4. *Those using colab or jupyter which maintain outputs don`t need to provide output screenshot*
5. *Don't close your editor, I will check outputs and code.*
6. *Any case of cheating will be given zero marks.*

Section A: Multiple Choice Questions (MCQs) (1 Marks)

1. Which method is used to remove missing values from a DataFrame?

- A) `remove()`
- B) `dropna()`
- C) `delete()`
- D) `trim()`

2. What does `plt.xlabel("X-Axis")` do?

- A) Renames the Y-axis
- B) Labels the X-axis
- C) Titles the chart
- D) Plots a label on data points

3. Which of the following is used to split data into train/test sets?

- A) `sklearn.model_selection.split()`
- B) `train_test_split()`
- C) `split_data()`
- D) `data_split()`

4. What does `fit()` do in a model?

- A) Predicts labels
- B) Splits data
- C) Trains the model
- D) Evaluates accuracy

Student ID: _____

5. What is the output of `print(type([1, 2, 3]))`?

- A) `<class 'tuple'>`
- B) `<class 'list'>`
- C) `<class 'array'>`
- D) `<class 'dict'>`

Section B: Write program basic python (2 Marks)

Q1 create a 3x3 identity matrix using NumPy and print its transpose. (1 marks)

Q2. Plot a sine wave using Matplotlib for values from 0 to 2π.

Section C: AI Algorithms (2 Marks)

1. Using `sklearn.datasets.fetch_california_housing`, train a linear regression model and predict house prices. Plot the predicted vs actual values.

2. Load the wine dataset and train a classifier. Print the accuracy score , classification report and plot confusion metrics.